

Incident Response Report

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Cyber Security Internship – Week 3

1. Executive Summary

A simulated incident involving reconnaissance and unauthorized SSH login attempts was conducted in a VirtualBox lab using Kali Linux and Metasploitable 2. Nmap was used to identify exposed services and failed SSH authentication attempts were detected through system logs. Appropriate containment, analysis, and recommendations are documented.

2. Incident Description

Incident Name: Simulated Unauthorized SSH Login Attempt
Environment: Kali Linux (attacker), Metasploitable 2 (target), Oracle VirtualBox.
Detection Method: System authentication logs and Nmap reconnaissance.

3. Indicators of Compromise (IOCs)

IOC	Details
Source IP	192.168.56.104
Target IP	192.168.56.101
Service Targeted	SSH (Port 22)
Evidence	Failed password for root in auth.log
Reconnaissance	Nmap service/version scan

4. Detection and Response Timeline

- Started Metasploitable target VM.
- Performed Nmap service/version scan from Kali.
- Attempted SSH login using incorrect credentials.
- Authentication failures observed in system logs.
- SSH service reviewed and containment documented.

5. Containment and Eradication Actions

- Reviewed SSH service status.
- Stopped SSH service during testing.
- Recommended isolating the affected machine from the network.
- Recommended resetting credentials and applying security updates.

6. Root Cause Analysis

Root Cause: Exposed SSH service accepted connection attempts from the local network.

Impact: Unauthorized login attempts were detected but unsuccessful.

Failure Point: Lack of intrusion detection and exposed legacy SSH configuration.

7. Lessons Learned

- Monitor authentication logs regularly.
- Disable unused services.
- Use strong passwords and modern SSH configurations.
- Perform routine vulnerability assessments.

8. Recommendations

- Deploy IDS/IPS (Snort/Suricata).
- Restrict SSH access.
- Keep systems patched.
- Conduct regular log reviews and vulnerability scans.

9. Evidence (Screenshots)

Nmap scan showing open ports and services.

```
kali-linux (Snapshot 1) [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

kali@kali: ~
Session Actions Edit View Help

kali@kali:~$ nmap -sV 192.168.56.101
Starting Nmap 7.90 ( https://nmap.org ) at 2024-07-10 15:28 -0400
Nmap scan report for 192.168.56.101
Host is up (0.82s latency).
Not shown: 578 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.3.4
23/tcp    open  telnet   Linux telnetd
25/tcp    open  smtp     Postfix smtpd
33/tcp    open  domain   ISC BIND 9.4.2
80/tcp    open  http     Apache httpd 2.2.8 ((Ubuntu) DAV/2)
112/tcp   open  rpcbind  2 (RPC #100000)
139/tcp   open  netbios-ssn Samba smbd 3.0 - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.0 - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec     netkit-rsh rshcd
513/tcp   open  login    netkit-rsh rlogind
514/tcp   open  shell    Netkit rshd
1800/tcp  open  java-rmi GNU Classpath gmiRegistry
1524/tcp  open  bindshell Metasploitable root shell
2800/tcp  open  rfs      2-4 (RPC #1000003)
2122/tcp  open  ftp      ProFTPD 1.3.1
3306/tcp  open  mysql    MySQL 5.0.51a-Jubuntu5
5432/tcp  open  postgresql PostgreSQL DB 8.2.0 - 8.3.7
5900/tcp  open  vnc      VNC (protocol 3.3)
6800/tcp  open  X11      (access denied)
6667/tcp  open  irc      Unreal3DC
8080/tcp  open  ajp13    Apache Jserv (Protocol v1.3)
8180/tcp  open  http     Apache/Ubuntu/Coyote JSP engine 1.1
MAC Address: 08:00:27:C9:FA:D3 (Oracle VM VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN, OS
s: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 14.15 seconds

kali@kali:~$
```

Authentication log showing failed SSH login attempts.



```
Metasploitable2 1 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

d for user root by (uid=0)
Jul 10 15:09:01 metasploitable CRON[4806]: pam_unix(cron:session): session close
d for user root
Jul 10 15:17:01 metasploitable CRON[4835]: pam_unix(cron:session): session opene
d for user root by (uid=0)
Jul 10 15:17:01 metasploitable CRON[4835]: pam_unix(cron:session): session close
d for user root
Jul 10 15:19:14 metasploitable sudo: msfadmin : TTY=ttty1 ; PWD=/home/msfadmin ;
USER=root ; COMMAND=/etc/init.d/ssh start
Jul 10 15:19:14 metasploitable sudo: pam_unix(sudo:session): session opened for
user root by msfadmin(uid=0)
Jul 10 15:19:14 metasploitable sudo: pam_unix(sudo:session): session closed for
user root
Jul 10 15:19:14 metasploitable sshd[4853]: Server listening on :: port 22.
Jul 10 15:19:14 metasploitable sshd[4853]: error: Bind to port 22 on 0.0.0.0 fai
led: Address already in use.
Jul 10 15:20:26 metasploitable sshd[4860]: pam_unix(sshd:auth): authentication f
ailure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.56.104 user=root
Jul 10 15:20:28 metasploitable sshd[4860]: Failed password for root from 192.168
.56.104 port 57564 ssh2
Jul 10 15:20:31 metasploitable sshd[4860]: Failed password for root from 192.168
.56.104 port 57564 ssh2
Jul 10 15:20:49 metasploitable sshd[4860]: PAM 1 more authentication failure; lo
gname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.56.104 user=root
msfadmin@metasploitable:~$ _
```

Failed SSH authentication from Kali Linux.

```
(kali㉿kali)-[~]  
$ ssh -o HostKeyAlgorithms=+ssh-rsa -o PubkeyAcceptedAlgorithms=+ssh-rsa root@192.168.56.101  
** WARNING: connection is not using a post-quantum key exchange algorithm.  
** This session may be vulnerable to "store now, decrypt later" attacks.  
** The server may need to be upgraded. See https://openssh.com/pq.html  
root@192.168.56.101's password:  
Permission denied, please try again.  
root@192.168.56.101's password:  
Permission denied, please try again.  
root@192.168.56.101's password: █
```